

SUPERCoolANT

DRY ICE

Frozen carbon dioxide gas is called dry ice, because it does not melt into a liquid when heated, but changes directly into a gas in a process known as sublimation. Dry ice is very cold, with a temperature as low as -109.3°F (-78.5°C). This property makes it very useful for keeping things cool, especially in the storage and shipping of food, medicine, and other supplies that spoil easily. When it becomes a gas, dry ice can be used to cool objects without wetting them and is also used to create fun special effects.

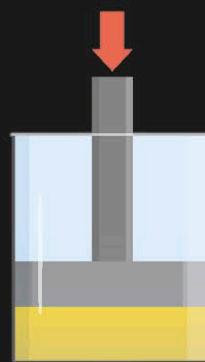
Dry ice pellets must be handled with insulated gloves to avoid frostbite or cold burns.

Cooling car cockpits

Formula 1 cars have powerful engines that produce a lot of heat. During a race, air cools the moving car, but if the engine is running while the car is not moving, the cockpit becomes very hot for the driver. Dry ice is sometimes used to cool the cockpit. Here, cold carbon dioxide that has sublimated from dry ice is being pumped into the cockpit of British driver Jenson Button's car at the 2011 Singapore Grand Prix.

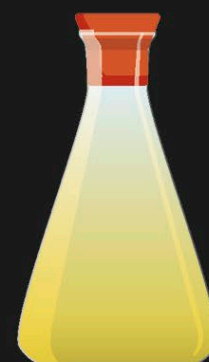
PROPERTIES OF GASES

Gases, liquids, and solids are the three different states of matter (see page 25). Each state has different properties. In gases, the molecules are far apart and free to move around. Gases are also very sensitive to pressure and temperature. A gas such as carbon dioxide can be made into solid dry ice when high pressure is applied to it, squeezing the space between the gas molecules and tightly packing them together.



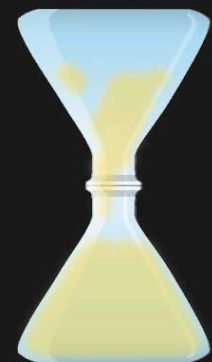
Easy to compress

The molecules in a gas move quickly in all directions. They can be forced into a smaller space.



Expand to fill up space

The molecules are randomly arranged and will expand to fill any space they are in.



Free-moving particles

The molecules in a gas have higher energy than those in a liquid or solid and move freely.